

Independent T4 Assembly Audit Record

ERDOZ81 — Paper I — Finite Fractional Split-Graph Theorem

Assembly target: **ERDOZ81-P1-ASSEMBLY-01**

Assembly claim. The admitted component gates combine, simultaneously and without circularity, to prove that for every finite split graph G on n vertices, $|E(G)| - 2v_3^*(G) \leq n^2/6 + n$.

Method. This is an assembly audit: components were treated as admitted at recorded statuses but every interface, notation mapping, loss charge, branch, and the exact final derivation were re-verified independently. All active-path component proofs were confirmed byte-identical to versions this auditor had already verified from scratch in prior sessions, and the assembled theorem was re-tested by an independent fractional-packing LP over random split graphs.

1. Package-Completeness Result

PACKAGE_COMPLETE. All 27 mandatory files match size and SHA-256 exactly against PACKAGE_MANIFEST.md; the archive matches its supplied .sha256 sidecar (2319db8b...5a4b33e8). README_FIRST.md, AUDIT_INSTRUCTIONS.md, and PACKAGE_MANIFEST.md are all present — the governance gap noted in the prior fractional-theorem audit is resolved in this bundle.

2. Independence Disclosure

The audit used only the supplied bundle and independently reconstructed each interface. The bundle embeds this auditor's own prior audit PDFs (BOUND-01, FRAC-THM-01) and the component gates' self-audit statuses; these were **not** used as authority. Component correctness rests on independent re-derivation (by hand and computation), not on the embedded verdicts.

3. Authoritative-Version Table

Active-path proofs verified byte-identical to independently-verified versions. Superseded versions present in the manifest (FRAC-THM v0.1/v0.2, PHASE2-DUAL v0.1) are excluded from the active path.

Component gate	Ver.	Status (recorded)	Active proof / check
DEF-01	0.1	PASS / CLEAN	well-definedness — VERIFIED
LOAD-01	0.1	PASS / CLEAN	$\Sigma \kappa_e = b_{\geq 2}/2$ — VERIFIED
PHASE1-01	0.1	PASS / CLEAN	w_1 feasible, v_1 exact — VERIFIED
LP-DUALITY-01	0.2	PASS / CLEAN	internal Farkas strong duality — VERIFIED
PHASE2-DUAL-01	0.2	PASS / CLEAN *	$v_{\text{con}} = b_{\geq 2}/2 + M(\kappa)$ — VERIFIED
CONCAVE-01	0.1	PASS / CLEAN	concavity direction — VERIFIED
PROFILE-01	0.1	PASS / CLEAN	pure-profile dominance — VERIFIED
Q0-01	0.1	PASS / CLEAN	$q=0$ branch — VERIFIED
SYMMETRY-01	0.1	PASS / CLEAN	orbit averaging — VERIFIED
3ORBIT-01	0.1	PASS / CLEAN	orbit-program equivalence — VERIFIED
3ORBIT-SOLVE-01	0.1	PASS / CLEAN	exact orbit optimum — VERIFIED
BOUND-01	0.1	PASS / CLEAN / IND-AUDITED	$M(\kappa) \geq R - p/2$ — VERIFIED
FRAC-THM-01	0.2 (proof v0.4)	PASS / CLEAN / IND-AUDITED	end-to-end assembly — VERIFIED

* PHASE2-DUAL-01 header records PASS/CLEAN, but its body still reads “evidence remains CANDIDATE / no independent audit” with no PASS event in its status history — a metadata inconsistency (see §10). Its mathematics is nonetheless independently verified.

4. Hypothesis-Compatibility Check

All components share one fixed split presentation $V(G)=K \cup I$ with $p=|K|$, $I=I_{\geq 2} \cup I_1 \cup I_0$, $q=|I_{\geq 2}|$, the same load vector κ , the same fixed clique polytope P_p , and the same functional $M(\kappa)=\min_{z \in P_p} \Sigma (1-\kappa_e)z_e$. The κ produced by LOAD/PHASE

(graph side) is exactly the aggregate profile $\Sigma n_s a^s$ consumed by PROFILE/BOUND (LP side); coordinates, domains, and the baseline $b=1$ match. **Compatible.**

5. Order-of-Application Check

DEF → LOAD → PHASE1 → LP-DUALITY → PHASE2-DUAL → (CONCAVE, PROFILE) → Q0 → (SYMMETRY, 3ORBIT, 3ORBIT-SOLVE) → BOUND → FRAC-THM. Each step's inputs are established strictly by earlier steps; the deficit identity is produced before the bound is inserted; the bound is inserted before the conversion to n . **Valid.**

6. Resource / Loss Ledger

The deficit identity $|E(G)| - 2V_{\text{com}} = C(p,2) - 2M(\kappa) + b_1$ is **exact** (the entire $b_{\geq 2}$ contribution cancels; verified on every test graph). The only inequality losses are: (i) the quantitative allowance $p/2$, charged once at BOUND-01 as $M(\kappa) \geq R - p/2$ [symbolically, $-2(R - p/2) = -2R + p$, appearing once]; (ii) the degree-one term b_1 , from the exact edge count, once. They are combined once via $p/2 + b_1 \leq n$, and the quadratic is enlarged once via $p+q \leq n$. No term is charged twice; no asymptotic, exceptional-set, parity, or numerical loss appears. The stronger U-branch loss $p/6$ is not double-counted. **Ledger consistent.**

7. Exceptional-Regime Check

All branches are covered exactly once and were confirmed computationally: **q=0** (Q0-01, no division by q ; $M(0)=p(p-1)/6$ for $p \geq 3$, ≤ 0 slack for $p \leq 2$); **q>0** (convex decomposition into pure profiles); **s=2** (SSS constraint omitted); **o=0,1,2,≥3** (only existing triangle types); **p<3** (no clique triangles; degenerate convention); **I₀** (enters only through n); **I₁** (exactly b_1 edges, no triangle load). No vertex class overlaps. **Complete.**

8. Circularity Check

The dependency graph is acyclic: definitions → packing construction & duality → $M(\kappa)$ → pure-profile bound → finite theorem. No component uses FRAC-THM-01 or ASSEMBLY-01 to establish an upstream claim; BOUND-01 does not depend on the packing construction, and the LP-duality gate is self-contained (internal Farkas, no external theorem). **No cycle.**

9. Exact Final Derivation

$V_{\text{com}} = b_{\geq 2}/2 + M(\kappa)$ (feasible packing) $\Rightarrow |E(G)| - 2V_{\text{com}} = C(p,2) - 2M(\kappa) + b_1$. With $M(\kappa) \geq R - p/2$ and the exact identity $C(p,2) - 2R + p = (p+q)^2/6 + p/2$ (confirmed in symbolic algebra, residual 0), $|E(G)| - 2V_{\text{com}} \leq (p+q)^2/6 + p/2 + b_1 \leq n^2/6 + n$. Since $V_{\text{com}} \leq v_3^*(G)$, the target follows. An independent fractional-packing LP over 800 fresh random split graphs (and 3,000 in the prior session) reproduced the bound with zero failures; the only equality case is the trivial empty graph. **Exact match with target.**

10. Material Objections and Exclusions

Material objections (assembly correctness): None. No incompatible hypothesis, stale version in the active path, omitted branch, logical cycle, duplicated loss, missing proof-critical dependency, over-strong conclusion, or packing/optimum mismatch was found.

Minor observations (documentation / governance, non-blocking):

- **Status-metadata inconsistency.** PHASE2-DUAL-01's record header reads PASS/CLEAN while its body says "evidence remains CANDIDATE / no independent audit" and its status history records no PASS promotion. The same header-vs-body pattern affects several self-audited T2 gates. Mathematically all are verified; the records should be synchronized before a freeze.
- **Dangling assembly dependency.** FRAC-THM-01 lists ERDOZ81-P1-BOUND-FALSIFY-01 as an ASSEMBLY_DEPENDENCY, but that gate is neither a component in the assembly table nor included in the bundle. It is a boundary-falsification support artifact and not proof-critical, but the reference should be reconciled.
- **Cosmetic.** The BOUND-01 gate record has a LaTeX rendering slip ("rac" for \frac) and DEF-C1 an "inom" typo —

display-only, no mathematical effect.

Exclusions / scope. This certifies only the finite fractional theorem for split graphs. It does not certify any fractional→integral transfer, asymptotic/general-graph extension, peer review, or a freeze; LIT-01 is citation-only and not proof-critical.

11. Final Verdict

RECOMMEND_PASS — ERDOZ81-P1-ASSEMBLY-01

The package is complete and the T4 assembly is independently verified without material objection. Authoritative component versions are correctly pinned and byte-identical to independently-verified proofs; hypotheses, application order, regime coverage, loss accounting, and circularity all check out; and the exact final derivation matches the target theorem $|E(G)| - 2v_3^*(G) \leq n^2/6 + n$, corroborated computationally. The two documentation items in §10 are non-mathematical and should be reconciled before any freeze/final-pass step, but they do not affect the correctness of the assembly.